Plotting functions with GnuPlot in 2D

Contents

Lab 1 - Plotting Functions 2
1.1 Intialisation 2
1.2 Plotting basic curves 2
1.3 Setting range and axis 3
1.4 Adding colours and changing line widths 4
1.5 Other features 5
Lab 2 - More gnuplot 5
2.1 Entering points to a data block and plotting 5
2.2 Loading code from a file 6
2.3 Setting titles and labels 6
2.4 Adjusting Sampling in a Range 7

Author: Koushani Roy
Roll No: 134
Semester: 1
Paper Code: C1PH230112P
Subject: Clab - 1, UG (Physics)
Institution: St. Xavier's College (Autonomous), Kolkata

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Lab 1 - Plotting Functions

1.1 Intialisation

Windows Type `wgnuplot` in desired directory in the terminal window.

Mac OS/Linux Does not have any dedicated GUI like `wgnuplot.exe` for Windows. Type `gnuplot` in termnal to begin.

Initialising in a Windows machine:

```
G N U P L O T
       Version 6.0 patchlevel 4    last modified 2025-12-18

       Copyright (C) 1986-1993, 1998, 2004, 2007-2025
       Thomas Williams, Colin Kelley and many others

       gnuplot home:     http://www.gnuplot.info
       faq, bugs, etc:   type "help FAQ"
       immediate help:   type "help"  (plot window: hit 'h')

       Terminal type is now windows
gnuplot>
```

Note: If mathematical symbols not rendered correctly, you must run `set term win enhanced` to be sure.

1.2 Plotting basic curves

```
       gnuplot> plot sin(1/x)
```

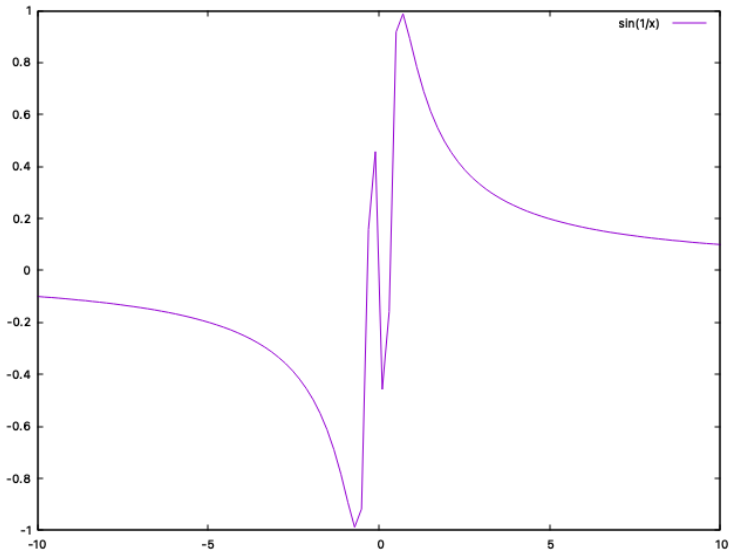


Figure 1: $\sin\left(\frac{1}{x}\right)$

1.3 Setting range and axis

```
gnuplot> set zeroaxis
gnuplot> plot [-pi:pi] sin(1/x**2)
```

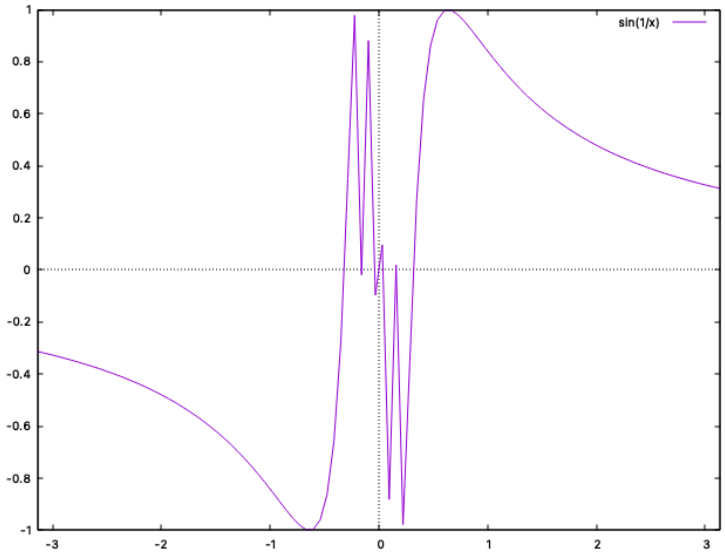


Figure 2: Range specified and zero axis made visible

1.4 Adding colours and changing line widths

```
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3, cos(x) ls
 ↪  7 lw 3
gnuplot> plot [-pi:pi] [:] sin(x) ls 2 lw 3, cos(x) ls
 ↪  7 lw 3 lc 'blue'
```

The part `[:]` lets the y-axis autoscale. You can stack subsequent functions in one graph by separating the instructions with a `,`.

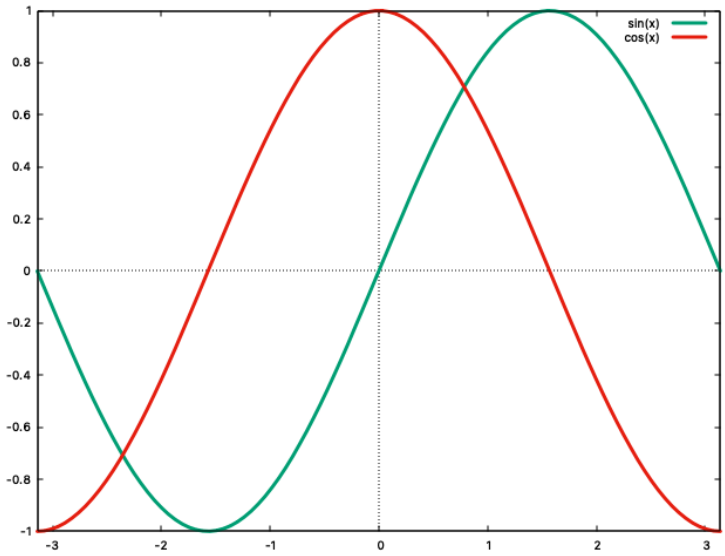


Figure 3: Default Linestyle 2 and 7 respectively for $sinx$ and $cosx$

`ls` stands for linestyle (one of the presets gnuplot comes with, different presets have different colour or dash pattern combos). However you can explicitely force features on the linestyles too for example, `lc 'blue'` forces linecolour blue on whatever colour linestyle 7 would have defaulted to.

`lw` stands for linewidth and determines how bold the lines are.

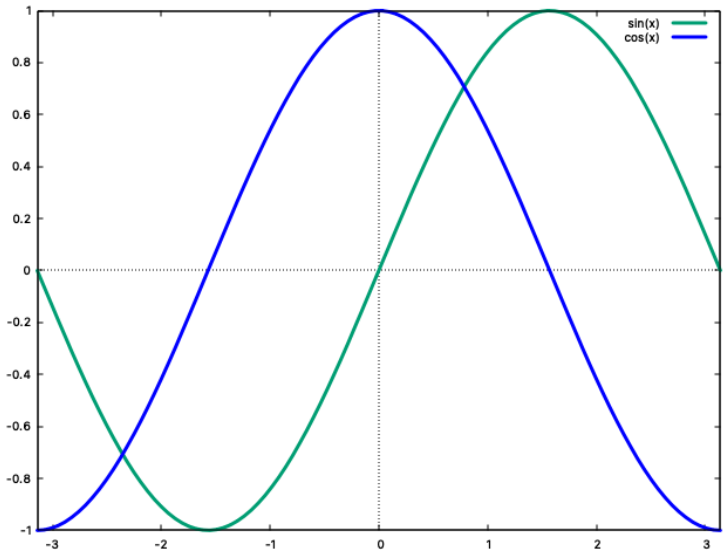


Figure 4: Forcing the colour blue on the linestyle 7 with def red

1.5 Other features

- There are different terminal types in gnuplot: some open an interactive window, some write straight to a file.
- Zoom in on functions by clicking and dragging a rectangle on the plot.
- Moving the cursor at different points on curve reveals the coordinates.
- Press 'r' to introduce a ruler element which is intersection of two lines and tells you the distance of different other points you hover from that given point.
- In mathematical operations use `**` for exponents.

```
gnuplot> h(x) = x**2 - 2**x + 1
gnuplot> plot h(x)
```

Lab 2 - More gnuplot

2.1 Entering points to a data block and plotting

The data block is named `$name_of_block` and a end-of-line character is specified with a << and data entry is taken per line, columns of two.

```
gnuplot> h(x) = sin(x)
gnuplot> $data << EOD
1.56 0.9998
```

```
2.34 0.567
EOD
gnuplot> ![alt text](image.png)
```

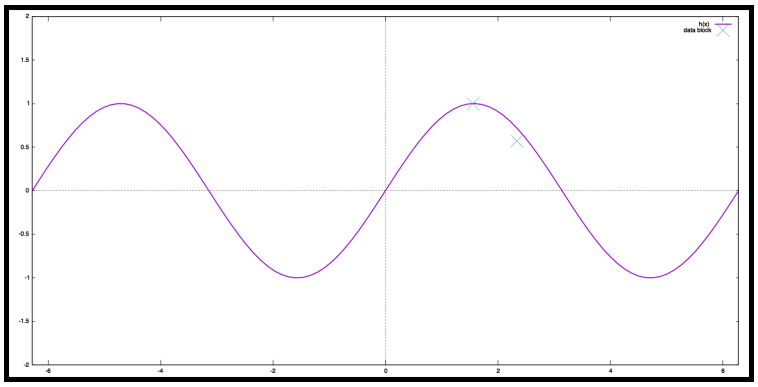


Figure 5: Crosshairs mark the data points

2.2 Loading code from a file

Provided you are in same directory as the file (quick check with `pwd`), it can be loaded with the `load` command. File extension is generally `txt` and it is customary to use `_plt`. In the prospective text file it is advised to use a `reset` command at beginning to restore default settings.

```
gnuplot> load 'mysine_plt.txt'
gnuplot> load 'mysine_plt.txt'
```

2.3 Setting titles and labels

The x and y axes names are customisable using `xlabel` and `ylabel` respectively. The function names can also be specified using `title`.

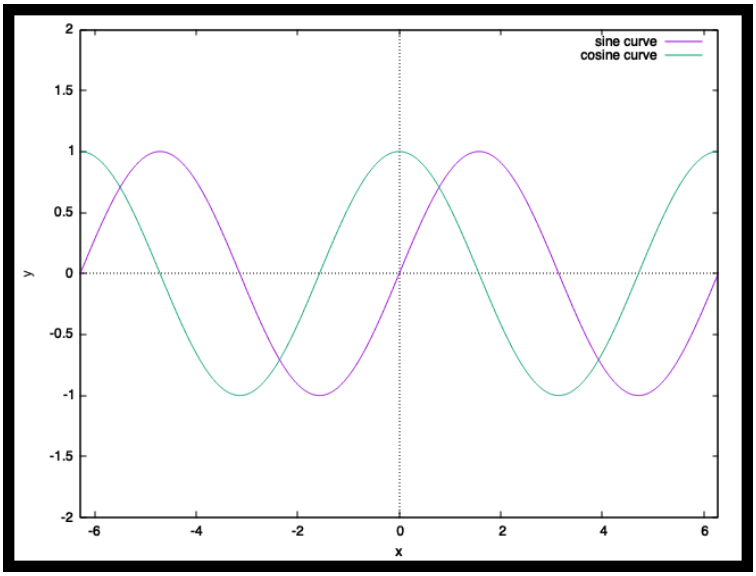


Figure 6: Naming figures

```
gnuplot> set xlabel 'x'
gnuplot> set ylabel 'y'
gnuplot> f(x) = sin(1/x)
gnuplot> g(x) = cos(1/x)
gnuplot> plot [-2*pi:2*pi] f(x) lw 2 title 'sine curve', g(x)
 ↪  lw 2 title 'cosine curve'
```

2.4 Adjusting Sampling in a Range

Increasing sample sizes makes curve more detailed, smoother.

```
gnuplot> show sample

        sampling rate is 100, 100 #default sampling size

gnuplot> plot sin(1/x)
gnuplot> set samples 1000
gnuplot> plot sin(1/x) #smoother curve
```

The default sampling size is 100.

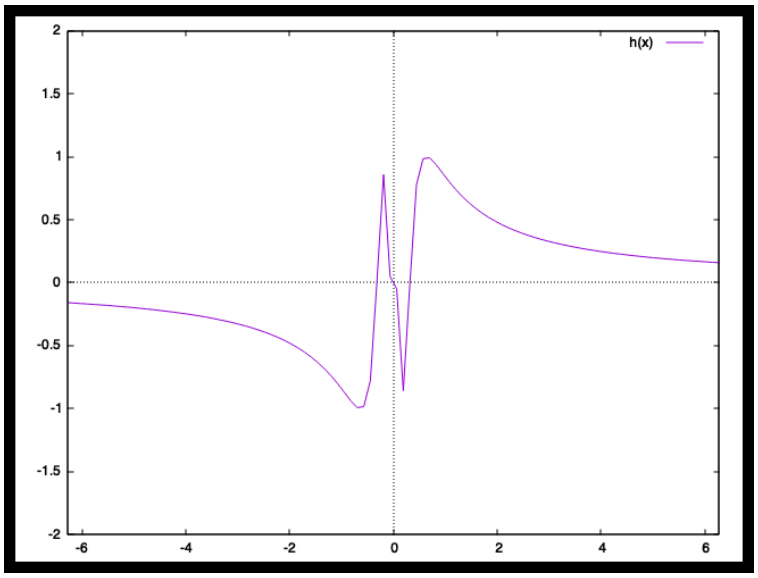


Figure 7: Sample size: 100 for $\sin\left(\frac{1}{x}\right)$

We set the sampling size to 1000 for a more detailed portion in the around zero range. Using `replot` instead of typing same cmd also works.

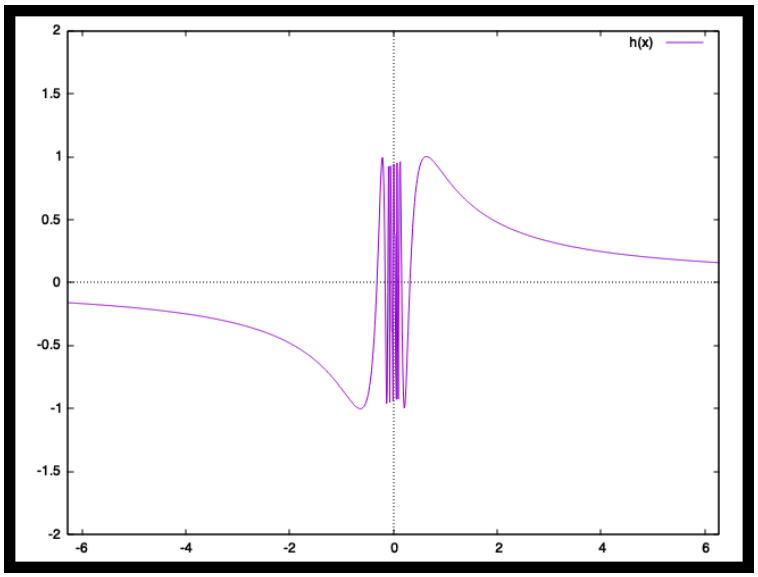


Figure 8: Sample size: 1000 for $\sin\left(\frac{1}{x}\right)$